



33-MAVZU · IV BOB. PYTHON, POSTGRESQL VA DJANGO

Django frameworkiga kirish. Arxitektura, loyiha va ilova (App) yaratish

Framework nima va nima uchun Django, MVT arxitekturasi, so'rovning Django ichidagi yo'li, o'rnatish, startproject va startapp, loyiha fayllari, settings.py, runserver, PostgreSQL'ga ulash.

8-sinf · Web Full-stack dasturlash

Bugun nimani o'rganamiz?

- **Framework** nima va nega **Django**
- **MVT** arxitekturası
- So'rov Django ichida qanday yo'l bosadi
- Loyiha (**project**) va ilova (**app**) yaratish
- Loyihani **PostgreSQL**'ga ulash

Framework nima?

- Veb-sayt uchun kerak: so'rovlar, URL'lar, HTML, formalar, login, parollar, admin panel... Noldan — **oylar!**

Framework — sinalgan tayyor "karkas": siz faqat loyihangizga xos qismni yozasiz



Uy qurish: noldan dastur — g'ishtni o'zingiz pishirish. Framework — tayyor karkas (poydevor, devor, quvurlar) ustida xonalarni bezash. Tezroq, xavfsizroq, sifatliroq.

Nima uchun Django?

-  **"Batareyalar bilan birga"** — admin panel, foydalanuvchilar, formalar, ORM tayyor
-  **Xavfsiz** — SQL injection, XSS, CSRF'dan avtomatik himoya
-  **Masshtablanadi** — Instagram, Pinterest, Spotify, Mozilla
-  Ajoyib hujjatlar, katta jamiyat

"Muddati bor perfeksionistlar uchun veb-framework"

MVT arxitekturası

Qism	Fayl	Vazifasi	Restoranda
Model	<code>models.py</code>	Ma'lumot va baza	Ombor
View	<code>views.py</code>	Mantiq: so'rov → javob	Oshpaz
Template	<code>templates/*.html</code>	Ko'rinish (HTML)	Likopcha va bezak
URL	<code>urls.py</code>	Manzil → View	Ofitsiant

- MVC'dagi Controller = Django'dagi **View**; MVC'dagi View = Django'dagi **Template**

So'rovning Django ichidagi yo'li

Matn

 Brauzer: GET /maqolalar/5/



 urls.py: "/maqolalar/5/" → maqola_view



 View: maqola_view(id=5) ⇔  Model: Maqola.objects.get(id=5) → PostgreSQL



▼ ma'lumot

 Template: maqola.html + ma'lumot



▼ tayyor HTML

 Brauzer

Loyiha va ilova

- **Loyiha (project)** — butun sayt: umumiy sozlamalar, asosiy URL'lar. **Bitta**
- **Ilova (app)** — mustaqil qism, bitta vazifa: `blog`, `accounts`, `shop`. **Bir nechta**



Maktab binosi: loyiha — bino (umumiy qoidalar, qo'ng'iroq). Ilovalar — xonalar: kutubxona, sport zali, oshxona. Har biri o'z ishini qiladi, lekin bitta binoda.

O'rnatish va loyiha yaratish

Terminal

```
mkdir maktab_blog && cd maktab_blog
python -m venv venv
venv\Scripts\activate
pip install django "psycopy[binary]" python-dotenv

django-admin startproject config .      # nuqta — joriy papkada!
python manage.py runserver              # http://127.0.0.1:8000 🚀
python manage.py startapp blog
```

config/settings.py

```
INSTALLED_APPS = [
    "django.contrib.admin",
    # ...
    "blog",          # ← ilovamizni ro'yxatdan o'tkazamiz
]
```

- **runserver** — faqat ishlab chiqish uchun; to'xtatish — **Ctrl+C**

Loyiha tuzilmasi

Matn

```
maktab_blog/
├── manage.py           ← boshqaruv buyruqlari
├── venv/               ← virtual muhit (GitHub'ga emas)
├── config/             ← LOYIHA
│   ├── settings.py    ← barcha sozlamalar
│   ├── urls.py        ← asosiy URL'lar
│   └── wsgi.py         ← serverga joylash (41-mavzu)
└── blog/              ← ILOVA
    ├── migrations/    ← baza o'zgarishlari tarixi
    ├── admin.py       ← admin panel (37)
    ├── models.py      ← modeller (36)
    └── views.py       ← view'lar (34)
```

manage.py buyruqlari

Buyruq	Vazifasi
<code>runserver</code>	Serverni ishga tushirish
<code>startapp nom</code>	Yangi ilova
<code>makemigrations</code>	Model o'zgarishlaridan migratsiya
<code>migrate</code>	Migratsiyalarni bazaga qo'llash
<code>createsuperuser</code>	Admin foydalanuvchi
<code>shell</code>	Loyiha ichida Python konsoli

settings.py — asosiy sozlamalar

Python

```
import os
from pathlib import Path
from dotenv import load_dotenv

BASE_DIR = Path(__file__).resolve().parent.parent
load_dotenv(BASE_DIR / ".env")

SECRET_KEY = os.getenv("SECRET_KEY") # MAXFIY!
DEBUG = os.getenv("DEBUG", "True") = "True" # serverda False
ALLOWED_HOSTS = os.getenv("ALLOWED_HOSTS", "127.0.0.1,localhost").split(",")

LANGUAGE_CODE = "uz"
TIME_ZONE = "Asia/Tashkent"
```



SECRET_KEY — GitHub'ga yuklanmaydi; **DEBUG = True** — faqat o'rganishda, internetda **doim False**

PostgreSQL'ga ulash

.env

```
SECRET_KEY=uzun-tasodifiy-satr
DEBUG=True
DB_NAME=maktab_blog
DB_USER=postgres
DB_PASSWORD=SIZNING_PAROLINGIZ
DB_HOST=localhost
DB_PORT=5432
```

settings.py

```
DATABASES = {
    "default": {
        "ENGINE": "django.db.backends.postgresql",
        "NAME": os.getenv("DB_NAME"),
        "USER": os.getenv("DB_USER"),
        "PASSWORD": os.getenv("DB_PASSWORD"),
        "HOST": os.getenv("DB_HOST"),
        "PORT": os.getenv("DB_PORT"),
    }
}
```

- Keyin: `python manage.py migrate` — Django o'z jadvallarini yaratadi (`auth_user`, `django_session` ...)

Birinchi sahifa

blog/views.py

```
from django.http import HttpResponse

def bosh_sahifa(request):
    return HttpResponse("Maktab blogiga xush kelibsiz! 🎉")
```

config/urls.py

```
from django.contrib import admin
from django.urls import path
from blog import views

urlpatterns = [
    path("admin/", admin.site.urls),
    path("", views.bosh_sahifa),      # bosh sahifa
]
```

- Bosh sahifani yangilang — raketa o'rnida sizning xabaringiz!

Amaliyot

- 1 venv → Django → `config` → `blog` → PostgreSQL → `migrate` → birinchi sahifa
- 2 `pip freeze > requirements.txt`
- 3 Git + GitHub (`.env` va `venv` yuklanmasin!)
- 4 Ataylab xato qiling (`HttpResponse`) — Django'ning **sariq xato sahifasini** o'qing
- 5 `python manage.py` — mavjud buyruqlar ro'yxati

```
.gitignore
venv/
.env
__pycache__/
*.pyc
db.sqlite3
```

Ko'p uchraydigan xatolar

- ❌ `No module named 'django'` — venv faol emas (terminalda `(venv)` yo'q)
- ❌ `startproject config` (nuqtasiz) — `config/config/` papkasi
- ❌ Ilova `INSTALLED_APPS` ga qo'shilmagan
- ❌ `That port is already in use` — server boshqa terminalda ishlayapti
- ❌ `password authentication failed` — `.env` paroli xato

Savol

Django'ning arxitekturalari qanday ataladi?

A MVC

B REST

C MVT (Model–View–Template)

D CRUD

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Ma'lumotlar tuzilmasi va baza bilan ishlash qaysi qismda?

A View

B Model

C Template

D URL

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Yaratilgan ilovani loyihaga qayerda ro'yxatdan o'tkazamiz?

A `urls.py`

B `views.py`

C `manage.py`

D `settings.py` → `INSTALLED_APPS`

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Internetdagi (production) saytda **DEBUG** qanday bo'lishi kerak?

A True

B False

C Ahamiyati yo'q

D None

Javobni tanlang — to'g'ri variant yashil bo'ladi



Uyga vazifa

- 1 Ikkinchi ilova `pages` — "Biz haqimizda" sahifasi (`/haqida/`)
- 2 MVT ni o'z hayotiy o'xshatishingiz bilan chizib tushuntiring (restoran emas!)
- 3 Django rasmiy qo'llanmasining 1-qismini o'qing

Yodda tuting

- Framework — tayyor karkas; Django — xavfsiz, "batareyalar bilan birga"
- **MVT**: Model — View — Template + URL
- `startproject config .` → `startapp blog` → `INSTALLED_APPS`
- `runserver` — ishlab chiqish serveri; `migrate` — jadvallar
- PostgreSQL: `DATABASES`; maxfiy qiymatlar — `.env`



Savollar bormi?

Keyingi mavzu: **34. Django URL va View komponentlari. URL va View yordamida sahifalar yaratish**

Batafsil matn, misollar va testlar: **33-django-kirish.html** darsida